

Data Structure (C Programming)

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Batch : B1

Assignment No 2

Structure with Functions.

1. Create a structure called "Student" with members name, age, and total marks.

i.

Create Function to display the details of students along with average: **void display(struct student s1)**

ii.

Create a function to find the average of each student and return the details of 1st ranker: **struct student ranker(struct student s1[])**

iii.

Create Function to read the details of student: **struct student Read()**

iv.

Write a C program to input data for 3 students using Read() and display their information using display() and find ranker using ranker() .

Input:

Assignment No: 2

- Que 1] Create a Structure called "Student" with members name, age and total marks.
- i. Create function to display the details of students along with average: `void display (struct student s1)`.
 - ii. Create a function to find the average of each student and return the details of 1st ranker: `struct student ranker (struct student s1[])`.
 - iii] Create function to read the details of student: `struct student Read()`.
 - iv] Write a C Program to input data for 3 students using `Read()` and display their information using `display()` and find ranker using `ranker()`.

display() → To print the student details.

ranker() → To find the first Rank

Read() → To Read the data.

```

→ #include <stdio.h>
struct student
{
    char name[50];
    int age;
    float total marks;
}

struct student Read()
{
    struct student s;
    printf("Enter Name:");
    scanf("%s", s.name);
    printf("Enter Age:");
    scanf("%d", &s.age);
    printf("Enter total marks:");
    scanf("%f", &s.total_marks);
    return s; // For Returning the student details.
}
    
```

```

void display(struct student s1)
{
    float average;
    average = s1.total_marks / 3;

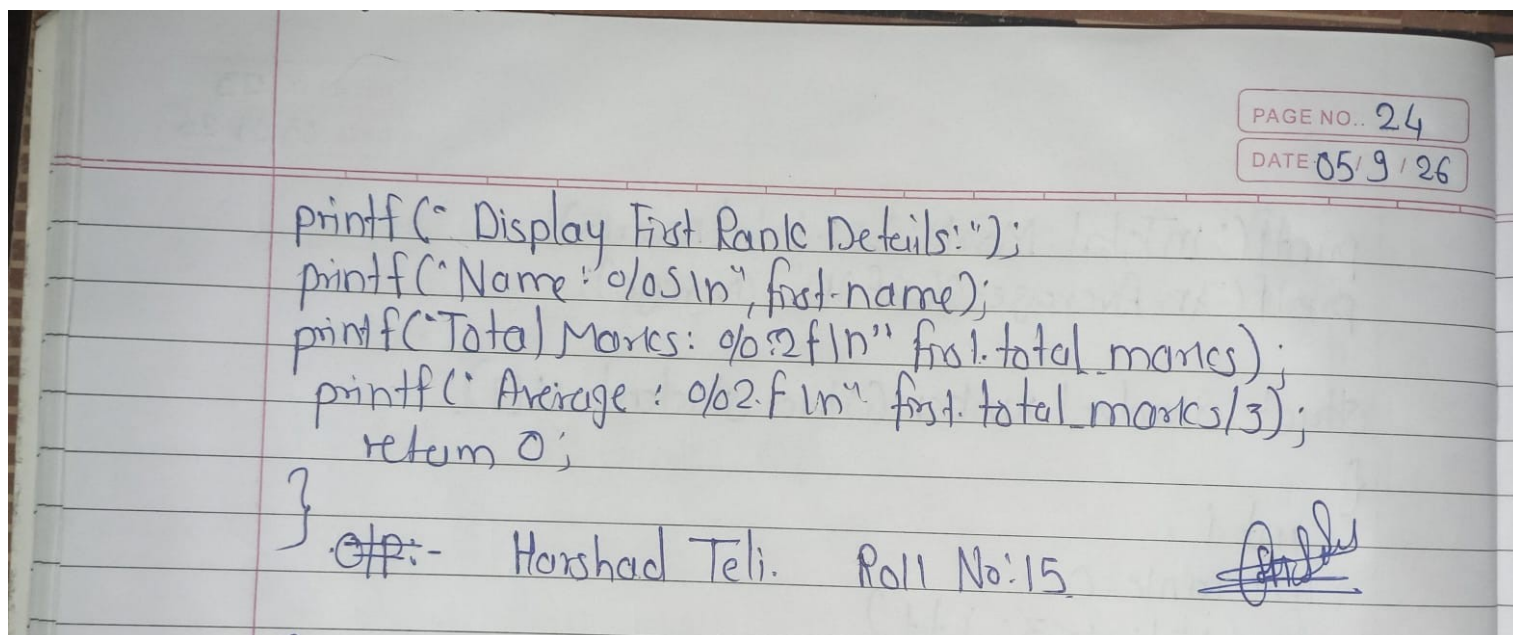
    printf("In Name: %s", s1.name);
    printf("In Age: %d", s1.age);
}
    
```

```
printf("In Total Marks: ", s1.total_marks);
printf("\n Average: %02f\n", average);
```

```
struct Student ranker (struct student s1[]).
```

```
{
    int i;
    int rank=0;
    for(i=1; i<3; i++)
    {
        if(s1[i].total_marks > s1[rank].total_marks)
        {
            rank=i;
        }
    }
    return s1[rank];
}
```

```
int main()
{
    struct student s[3];
    struct student first;
    int i;
    for(i=0; i<3; i++)
    {
        s[i] = Read();
    }
    printf("Student Details");
    for(i=0; i<3; i++)
    {
        printf("In Student %d", i+1);
        display(s[i]);
    }
    first = ranker(s);
}
```

Output:

```

student@student-OptiPlex-SFF-7020:~/Videos$ ./output

Enter details of Student 1
Enter name: Harshad
Enter age: 21
Enter total marks: 300

Enter details of Student 2
Enter name: MadhuShree
Enter age: 22
Enter total marks: 500

Enter details of Student 3
Enter name: Anuja
Enter age: 21
Enter total marks: 450

----- STUDENT DETAILS -----

Student 1
Name: Harshad
Age: 21
Total Marks: 300.00
Average: 100.00

Student 2
Name: MadhuShree
Age: 22
Total Marks: 500.00
Average: 166.67

Student 3
Name: Anuja
Age: 21
Total Marks: 450.00
Average: 150.00

----- FIRST RANK STUDENT -----
Name: MadhuShree
Age: 22
Total Marks: 500.00
Average: 166.67
student@student-OptiPlex-SFF-7020:~/Videos$

```